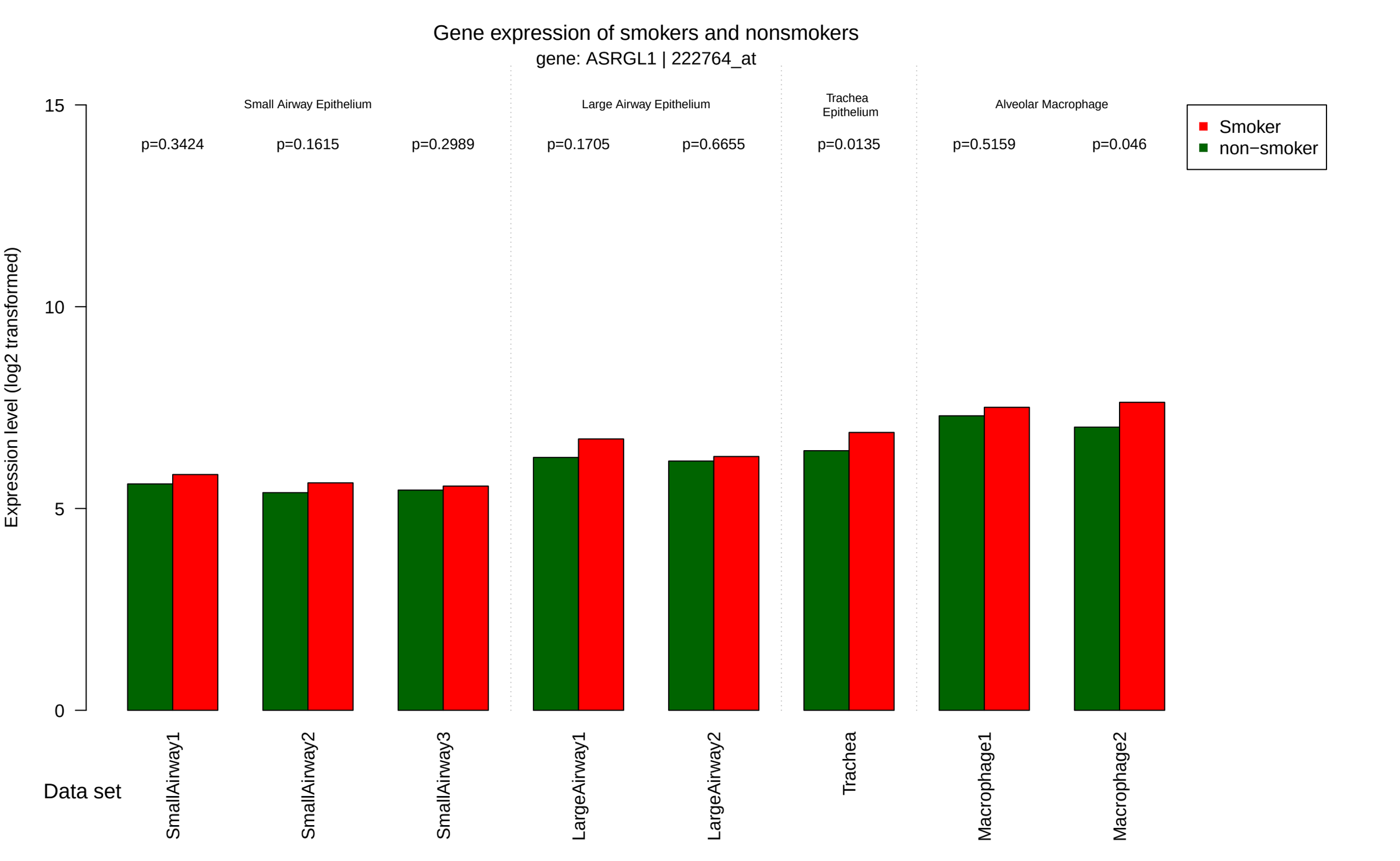
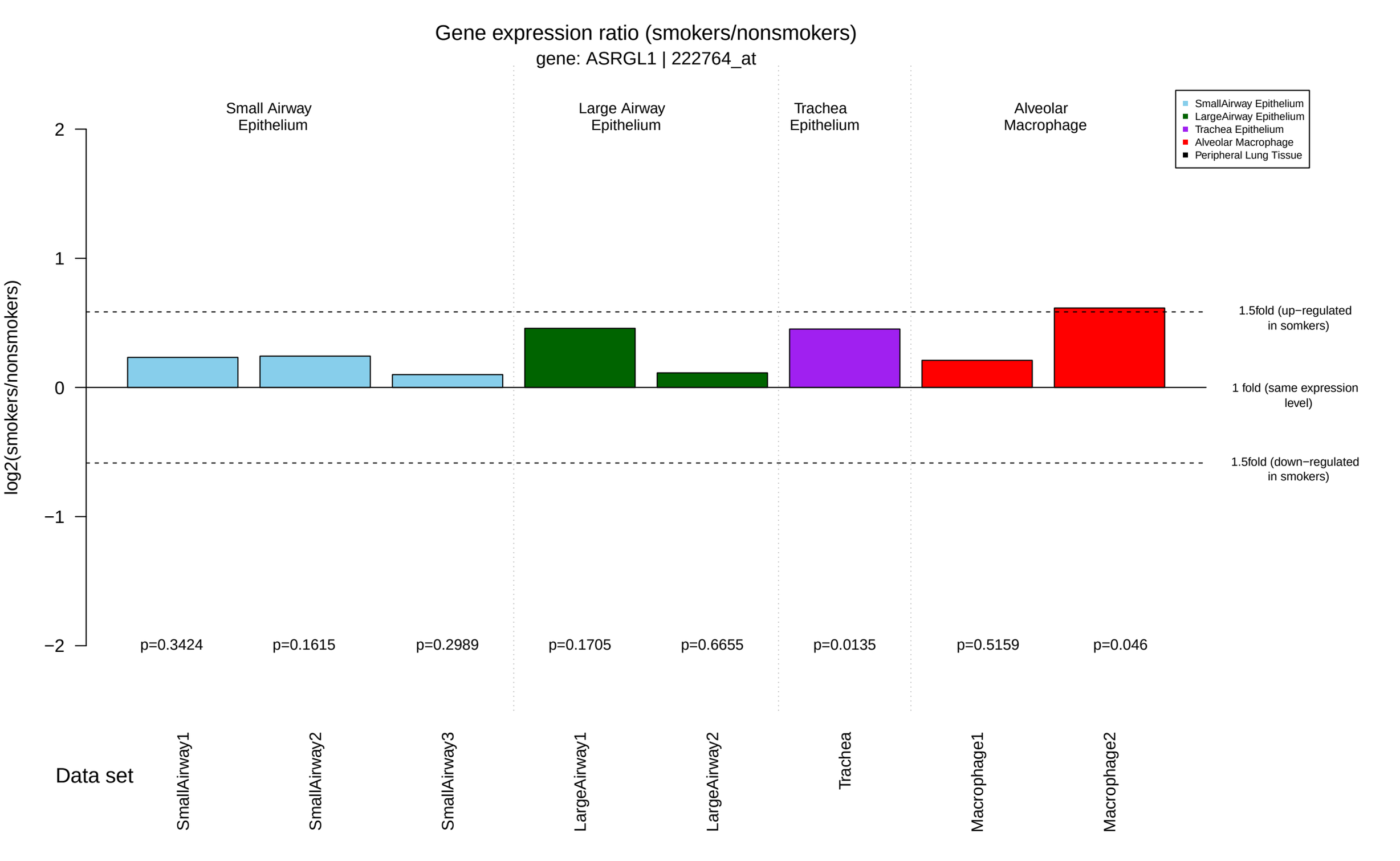
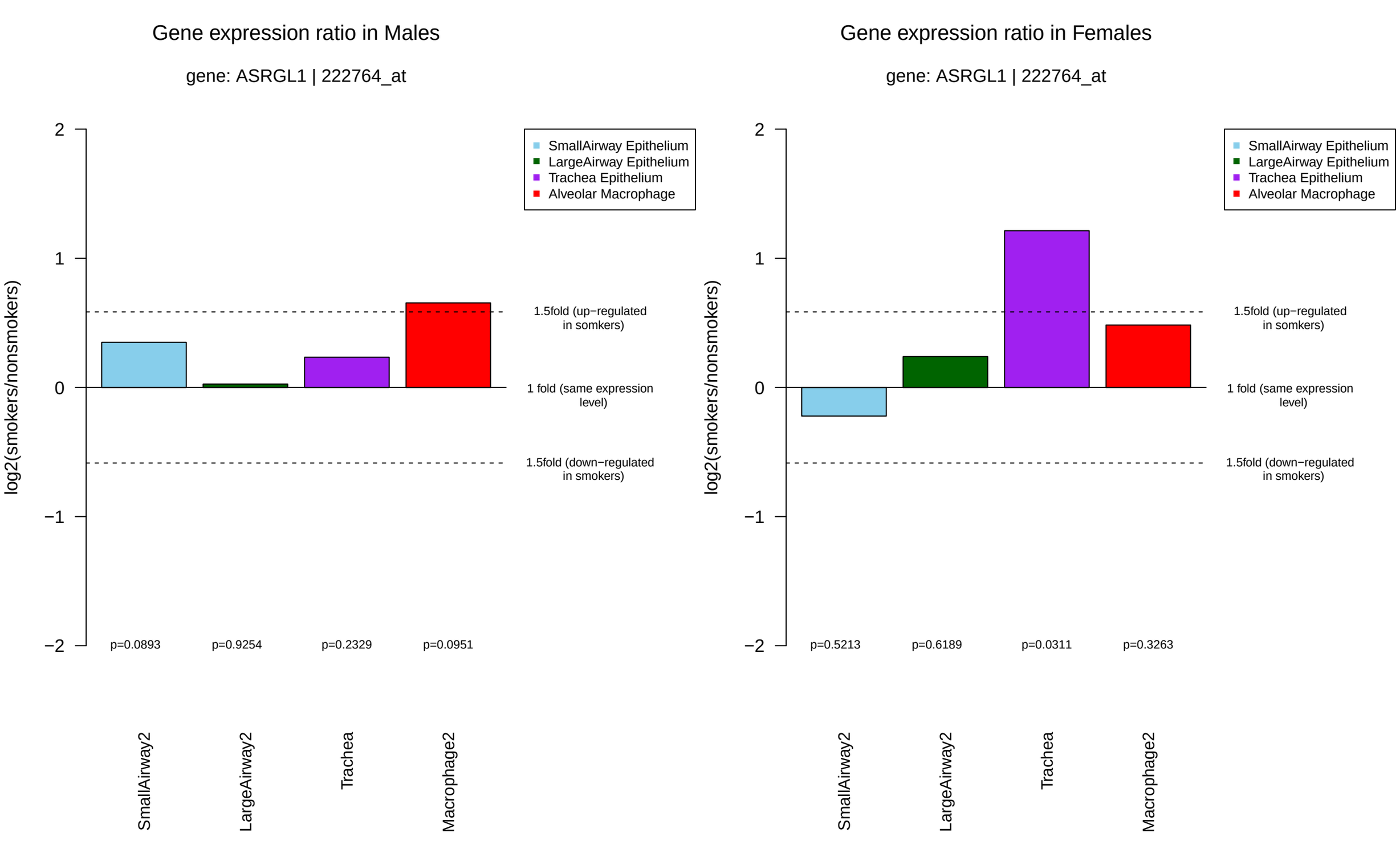
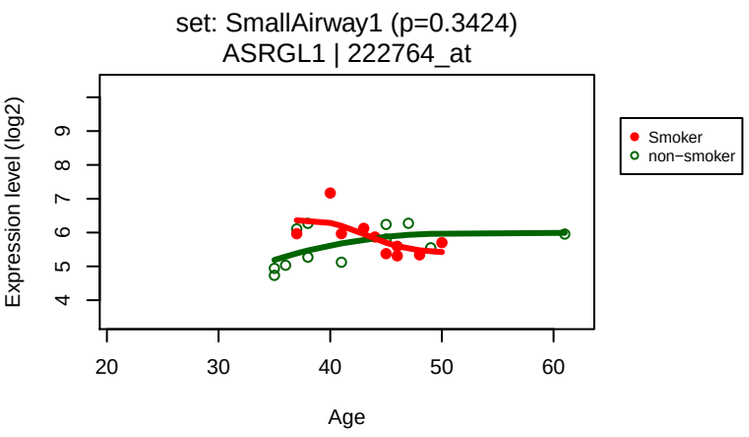
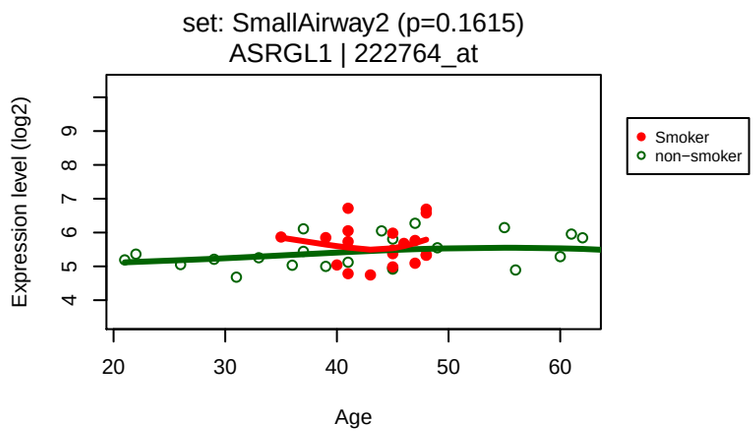
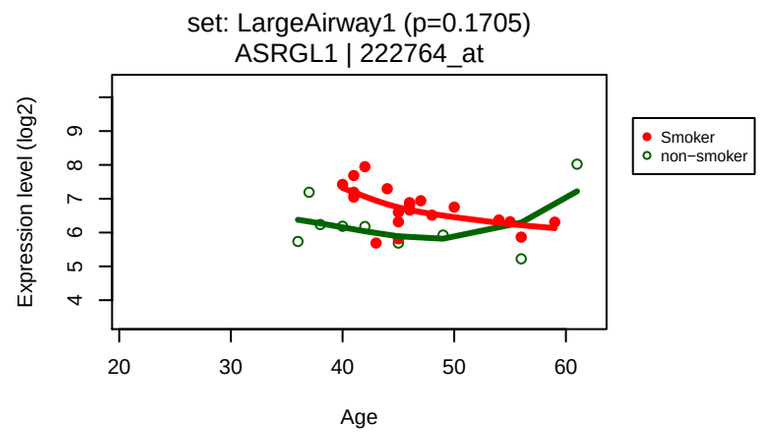

Gene expression of smokers and nonsmokers
gene: ASRGL1 | 222764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3424
p=0.1615
p=0.2989
p=0.1705
p=0.6655
p=0.0135
p=0.5159
p=0.046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

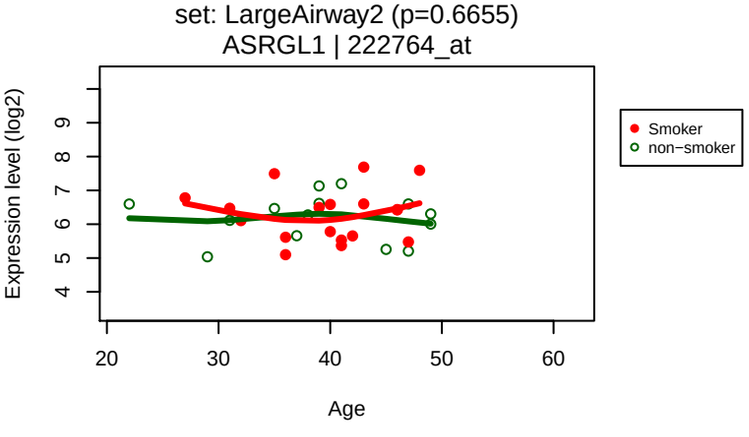

Gene expression ratio (smokers/nonsmokers)
gene: ASRGL1 | 222764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3424
p=0.1615
p=0.2989
p=0.1705
p=0.6655
p=0.0135
p=0.5159
p=0.046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

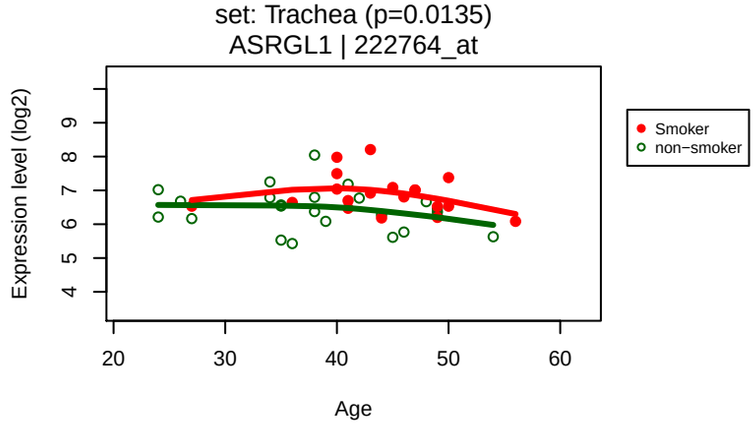

Gene expression ratio in Males
gene: ASRGL1 | 222764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0893
p=0.9254
p=0.2329
p=0.0951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASRGL1 | 222764_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5213
p=0.6189
p=0.0311
p=0.3263
SmallAirway2
LargeAirway2
Trachea
Macrophage2

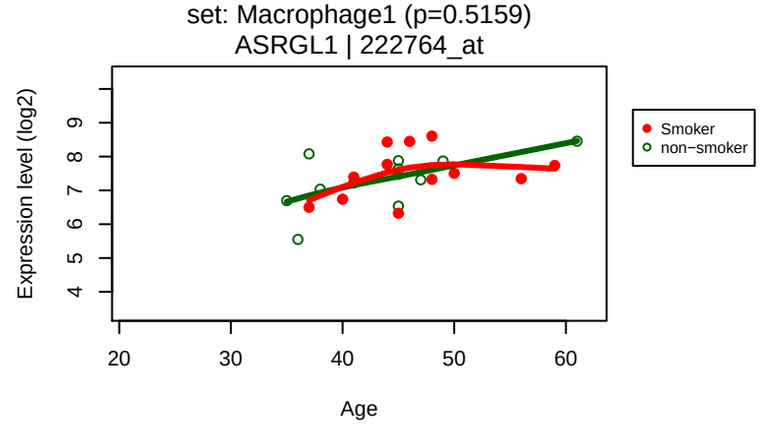

set: SmallAirway1 (p=0.3424)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker

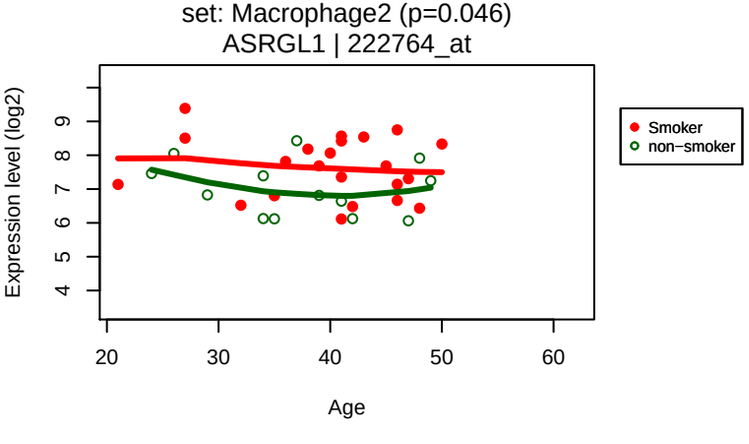

set: SmallAirway2 (p=0.1615)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker

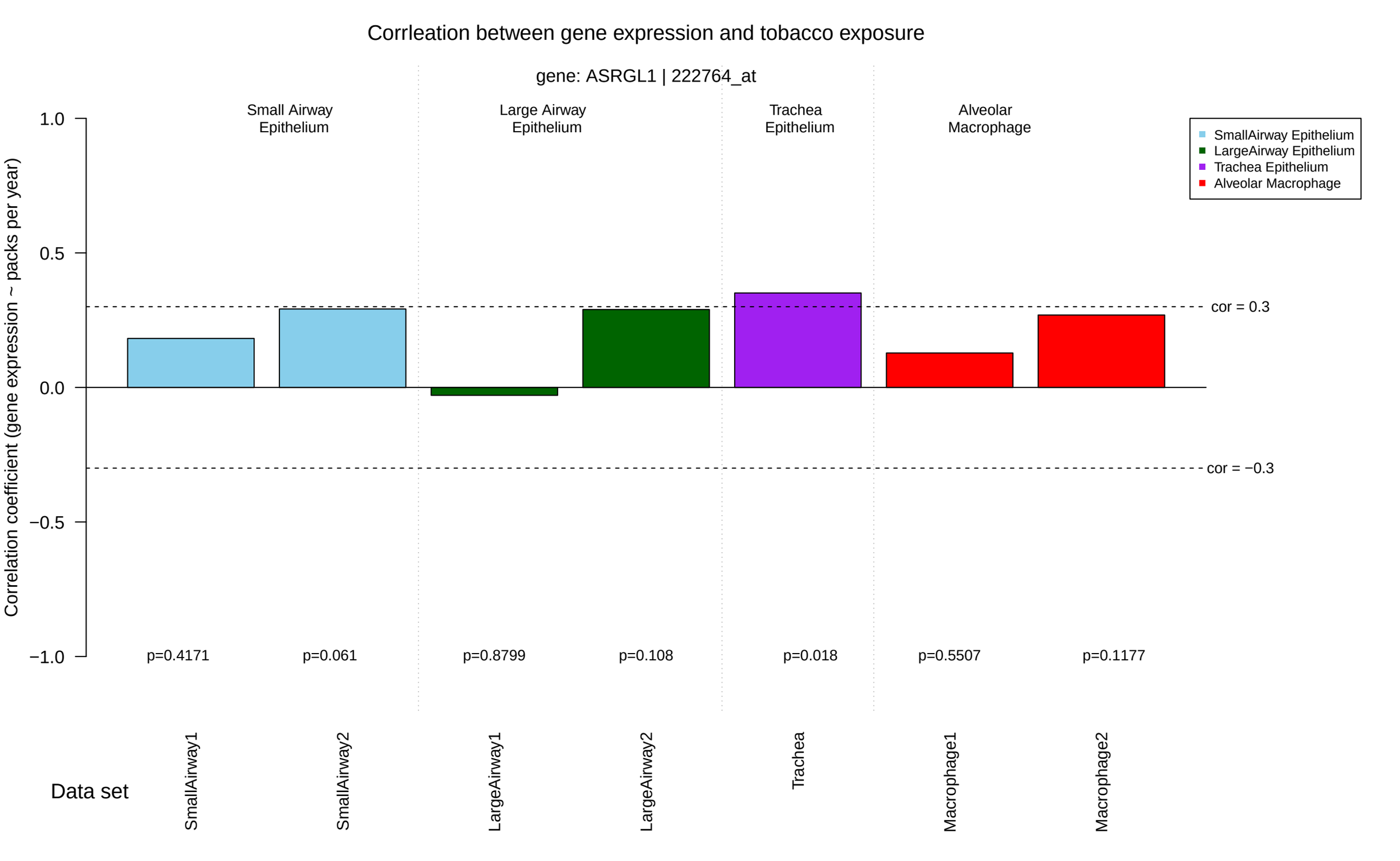

set: LargeAirway1 (p=0.1705)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6655)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0135)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5159)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.046)
ASRGL1 | 222764_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASRGL1 | 222764_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4171
p=0.061
p=0.8799
p=0.108
p=0.018
p=0.5507
p=0.1177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2